

Name- OM YADAV
SAP ID 590032772
Batch NO. 15

HDOC DAY-10 ACTIVITY REPORT: BATCHES 15 & 16 PART 1: TIME CONVERTER PROGRAM 1.1 Algorithm

1. Start

2. Input: Read total seconds (total_seconds).

3. Calculate Days: $\text{days} = \text{total_seconds} / 86400$

4. Calculate Remaining Seconds: $\text{remaining_seconds} = \text{total_seconds} \% 86400$

5. Calculate Hours: $\text{hours} = \text{remaining_seconds} / 3600$

6. Update Remaining Seconds: $\text{remaining_seconds} = \text{remaining_seconds} \% 3600$

7. Calculate Minutes: $\text{minutes} = \text{remaining_seconds} / 60$ 8.

Calculate Seconds: $\text{seconds} = \text{remaining_seconds} \% 60$

9. **Output:** Display days, hours, minutes, and seconds.

10. Stop

1.2 Pseudocode

text

BEGIN

Name- OM YADAV

SAP ID 590032772

Batch NO. 15

```
NUMBER total_seconds, days, hours, minutes, seconds
OUTPUT "Enter time in seconds: "
INPUT total_seconds

    days = total_seconds / 86400    hours =
(total_seconds % 86400) / 3600    minutes =
((total_seconds % 86400) % 3600) / 60    seconds
= ((total_seconds % 86400) % 3600) % 60

    OUTPUT days, "Days,", hours, "Hours,", minutes,
"Minutes,", seconds, "Seconds"
END
```

Use code with caution.

1.3 Flowchart Text Representation

text

```
[ Start ]
  |
  ▼
[ Input total_seconds ]
  |
  ▼
[ days = total_seconds / 86400 ]
[ rem_seconds = total_seconds % 86400 ]
  |
  ▼
[ hours = rem_seconds / 3600 ]
[ rem_seconds = rem_seconds % 3600 ]
  |
  ▼
[ minutes = rem_seconds / 60 ]
[ seconds = rem_seconds % 60 ]
```

Name- OM YADAV
SAP ID 590032772



[Print Days, Hours, Minutes, Seconds]



[Stop]

Batch NO. 15

Use code with caution.

1.4 Test Case Table

Test Case ID	Input	Expected Output	Statu
TC1_01	0	0 Days, 0 Hours, 0 Minutes, 0 Seconds	Pass
TC1_02	45	0 Days, 0 Hours, 0 Minutes, 45	Pass
TC1_03	3750	0 Days, 1 Hours, 2 Minutes, 30	Pass
TC1_04	90061	1 Days, 1 Hours, 1 Minutes, 1 Seconds	Pass

1.5 Source Code (time_converter.c)

Name- OM YADAV

SAP ID 590032772

Batch NO. 15

```
UW PICO 5.09 File
#include <stdio.h>

int main() {
    long long total_seconds;
    int days, hours, minutes, seconds;

    printf("Enter time in seconds: ");
    if (scanf("%lld", &total_seconds) != 1 || total_seconds < 0) {
        printf("Invalid input. Please enter a positive integer.\n");
        return 1;
    }

    days = total_seconds / 86400;
    int rem_seconds = total_seconds % 86400;

    hours = rem_seconds / 3600;
    rem_seconds = rem_seconds % 3600;

    minutes = rem_seconds / 60;
    seconds = rem_seconds % 60;

    printf("%lld seconds = %d Days, %d Hours, %d Minutes, %d Seconds\n",
        total_seconds, days, hours, minutes, seconds);

    return 0;
}
```

```
[omyadav@OMs-MacBook-Air ~ % clang day10.1.c -o day10.1
[omyadav@OMs-MacBook-Air ~ % ./day10.1
Enter a year: 2024
2024 is a leap year.
[omyadav@OMs-MacBook-Air ~ % ./day10.1
Enter a year: 1900
1900 is not a leap year.
[omyadav@OMs-MacBook-Air ~ % ./day10.1
Enter a year: 2000
2000 is a leap year.
[omyadav@OMs-MacBook-Air ~ % ./day10.1
Enter a year: 2023
2023 is not a leap year.
[omyadav@OMs-MacBook-Air ~ %
```

PART 2: LEAP YEAR CHECKER (COMPACT & LOGICAL OPTIMIZED) 2.1 Algorithm

1. Start

Name- OM YADAV

SAP ID 590032772

2. Input: Read year.

Name- ADITYA PRASAD

SAP ID 590032113

Batch NO. 15

3. Condition Check: If (year % 4 == 0 AND year % 100 != 0)
OR (year % 400 == 0), flag as Leap Year.

4. Else: Flag as Not a Leap Year.

5. Output: Print validation result.

6. Stop

2.2 Pseudocode

text

BEGIN

 NUMBER year

 OUTPUT "Enter a year: "

 INPUT year

 IF ((year % 4 == 0 AND year % 100 != 0) OR (year % 400
== 0)) THEN

 OUTPUT year, " is a leap year."

 ELSE

 OUTPUT year, " is not a leap year."

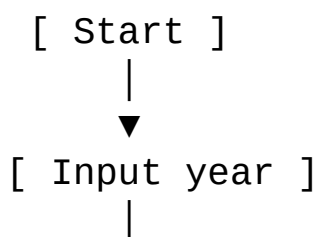
 ENDIF

END

Use code with caution.

2.3 Flowchart Text Representation

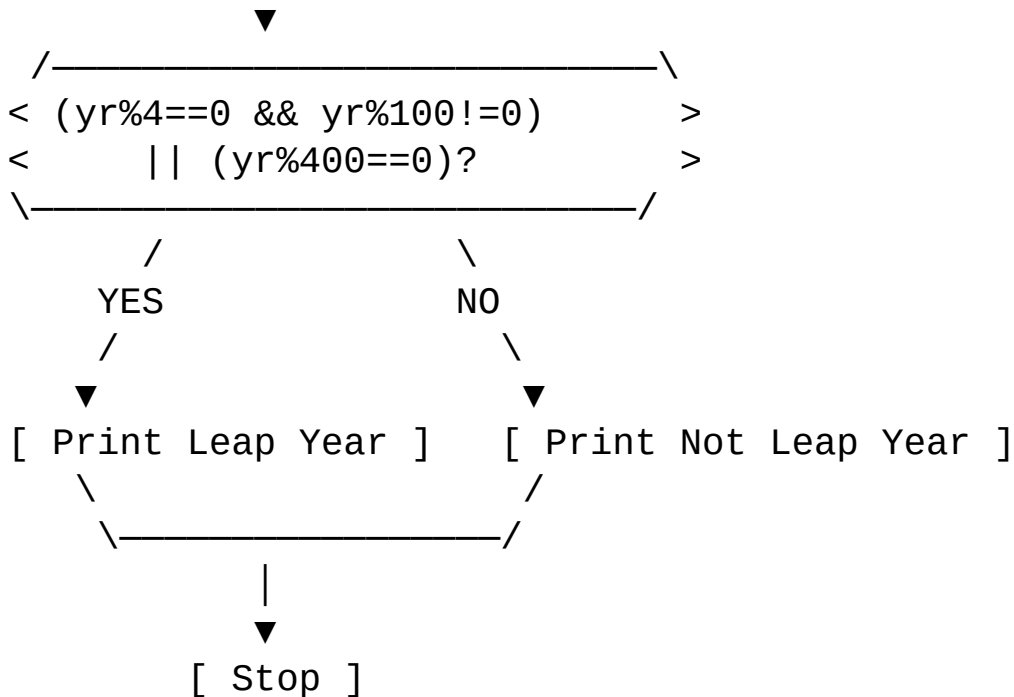
text



Name- ADITYA PRASAD

SAP ID 590032113

Batch NO. 15



Use code with caution.

2.4 Test Case Table

Test Case ID	Input (Year)	Expected Output	Status
TC2_01	2024	2024 is a leap year.	Pass
TC2_02	1900	1900 is not a leap year.	Pass
TC2_03	2000	2000 is a leap year.	Pass
TC2_04	2023	2023 is not a leap year.	Pass

Name- ADITYA PRASAD

SAP ID 590032113

Batch NO. 15

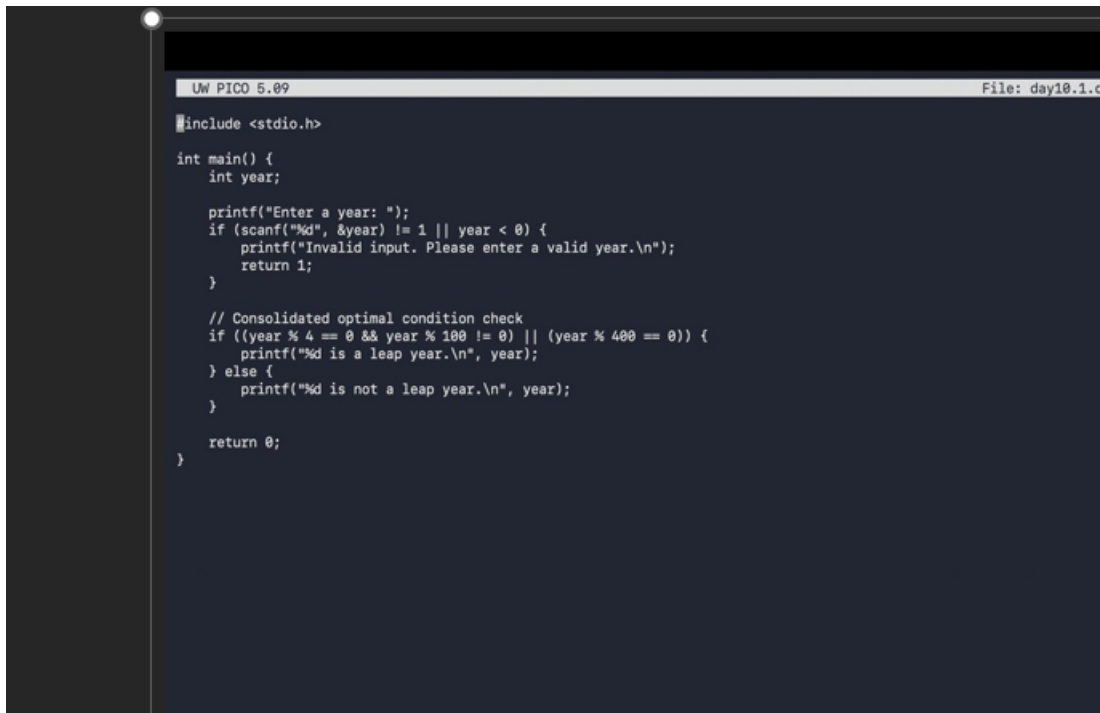
2.5 Source Code (leap_year.c)

C

Name- ADITYA PRASAD

SAP ID 590032113

Batch NO. 15



The image shows a terminal window with a dark background. At the top, there is a status bar with "UW PICO 5.09" on the left and "File: day10.1.c" on the right. The main area of the terminal displays a C program for checking leap years. The code includes a header file, a main function that prompts the user for a year, and logic to validate the input and determine if it's a leap year based on standard rules (divisible by 4 but not 100, or divisible by 400). The code is as follows:

```
#include <stdio.h>

int main() {
    int year;

    printf("Enter a year: ");
    if (scanf("%d", &year) != 1 || year < 0) {
        printf("Invalid input. Please enter a valid year.\n");
        return 1;
    }

    // Consolidated optimal condition check
    if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {
        printf("%d is a leap year.\n", year);
    } else {
        printf("%d is not a leap year.\n", year);
    }

    return 0;
}
```

Name- ADITYA PRASAD

SAP ID 590032113

Batch NO. 15

[Restored 3 Sep 2026 at 9:32:00 PM]

Last login: Thu Sep 3 16:32:05 on console

Restored session: Wed Sep 2 23:43:45 IST 2026

[omyadav@OMs-MacBook-Air ~ % nano day10.c

[omyadav@OMs-MacBook-Air ~ % clang day10.c -o day10

[omyadav@OMs-MacBook-Air ~ % ./day10

Enter time in seconds: 12

12 seconds = 0 Days, 0 Hours, 0 Minutes, 12 Seconds

[omyadav@OMs-MacBook-Air ~ % ./day10

Enter time in seconds: 34

34 seconds = 0 Days, 0 Hours, 0 Minutes, 34 Seconds

[omyadav@OMs-MacBook-Air ~ % ./day10

Enter time in seconds: 7

7 seconds = 0 Days, 0 Hours, 0 Minutes, 7 Seconds

[omyadav@OMs-MacBook-Air ~ % ./day10

Enter time in seconds: 99

99 seconds = 0 Days, 0 Hours, 1 Minutes, 39 Seconds

[omyadav@OMs-MacBook-Air ~ % nano day10.c

[omyadav@OMs-MacBook-Air ~ % clang day10.c -o day10

[omyadav@OMs-MacBook-Air ~ % ./day10

Enter time in seconds: 45

45 seconds = 0 Days, 0 Hours, 0 Minutes, 45 Seconds

[omyadav@OMs-MacBook-Air ~ % ./day10

Enter time in seconds: 3750

3750 seconds = 0 Days, 1 Hours, 2 Minutes, 30 Seconds

[omyadav@OMs-MacBook-Air ~ % ./day10

Enter time in seconds: 90061

90061 seconds = 1 Days, 1 Hours, 1 Minutes, 1 Seconds

omyadav@OMs-MacBook-Air ~ % █

PART 3: COMPILATION & EXECUTION INSTRUCTIONS

Name- OM YADAV
SAP ID 5900327721
Batch NO. 15

To complete the execution step required for Day-10,
students can run the following terminal sequences: **bash**

```
# Compile the programs
gcc time_converter.c -o time_converter
gcc leap_year.c -o leap_year
# Execute the programs
./time_converter
./leap_year
```